Material Behavior Prediction

Milestone 5 | 4 Models Evaluated

Complete Technical Guide to Milestone 5: Model Comparison & Metric Evaluation

1. Why Compare Multiple Algorithms?

No single machine learning model is universally best for every physical domain. In Milestone 4, we built a simple Linear Regression baseline. In Milestone 5, we test a structured progression of 4 algorithms to discover how non-linear tree structures and ensembling methods improve predictions:

2. The Final Model Comparison Table (Unseen Test Data)

Evaluated on the strictly held-out **201 test formulations**:

Model	Train R^2	Test R^2	Test MAE (MPa)	Test RMSE (MPa)	Generalization Gap	Key Takeaway
1. Linear Regression	0.6099	0.5802	8.90	11.19	0.0297	Underfits non-linear curves (Abrams' Law). High interpretability, moderate accuracy.
2. Decision Tree	0.8641	0.7885	5.95	7.94	0.0755	Captures physical thresholds (+21% R^2), but step functions lack smoothness.
3. Random Forest	0.9824	0.9069	3.55	5.27	0.0755	Averages 100 bagging trees; cuts error by 60% compared to baseline.
4. Gradient Boosting ★	**0.9699**	**0.9198**	**3.45**	**4.89**	**0.0501**	**TOP PERFORMER: Explains 92% of strength variance with lowest error.**

The Power of Ensembling in Materials Science

Notice the dramatic leap from Linear Regression to Gradient Boosting:

- **R^2 Score:** jumped from **58.0% → 92.0%** (+34% more variance explained).
- **Mean Absolute Error (MAE):** plummeted from **8.90 MPa → 3.45 MPa** (a **61.2% error reduction**!).
- **Root Mean Squared Error (RMSE):** dropped from **11.19 MPa → 4.89 MPa**.

3. Detailed Architectural & Scientific Analysis

1 Linear Regression (OLS Baseline)

Mechanism: Fits a flat 8-dimensional hyperplane to the data.

Why it plateaued at $R^2 \approx 58\%$: Concrete physics does not follow straight lines. The water-to-cement ratio relationship is an exponential inverse curve (Abrams' Law), and curing age follows a logarithmic curve. A single straight plane cannot bend to capture these simultaneous curvatures.

2 Decision Tree Regressor (`max_depth=6`)

Mechanism: Recursively splits features at optimal numerical thresholds (e.g. `age ≤ 21 days` , `cement ≤ 310 kg`).

Why it improved ($R^2 = 79\%$): It easily partitions data into distinct physical regimes (e.g., low water-cement mixes vs. high water-cement mixes). However, single decision trees produce discontinuous step functions, making fine-grained predictions slightly jagged.

3 Random Forest Regressor (100 Bagging Trees)

Mechanism: Trains 100 deep decision trees on random bootstrap subsets of data and averages their predictions (Bagging = Bootstrap Aggregating).

Why it excelled ($R^2 = 91\%$): By averaging 100 de-correlated trees, individual tree errors cancel out. Random Forest smooths the jagged step boundaries of single trees and eliminates sensitivity to outliers.

4 Gradient Boosting Regressor (100 Boosting Trees)

Mechanism: Builds trees sequentially. Tree 1 makes a baseline prediction; Tree 2 is trained to predict the *errors (residuals)* of Tree 1; Tree 3 predicts the remaining errors of Tree 2, and so forth.

Why it won ($R^2 = 92\%$, MAE = 3.45 MPa): Gradient Boosting acts as an iterative optimizer. In materials science, where subtle multi-component interactions exist (e.g., superplasticizer activating fine binders), boosting iteratively refines predictions for complex edge cases.

4. Overfitting & Generalization Diagnostics

In `visualizations/10_train_vs_test_r2_overfitting.png` , we examine the gap between Train R^2 and Test R^2:

- **Linear Regression:** Low train (0.61) and low test (0.58) → *High Bias (Underfitting)*.
- **Random Forest:** High train (0.98) and high test (0.91) → *Slight overfitting, but excellent generalization*.

- **Gradient Boosting:** Train (0.97) and test (0.92) with a tiny generalization gap of **0.05** → *Ideal balance between bias and variance*.